SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned checkpoint — 19 July 2026

Authority. Corrected French lines 2807–2819; original printed pp. 93–94; physical source-PDF p. 82; recomposed running p. 74. The printed-page-94 marker occurs at line 2813 immediately before "Indeed." Line 2820 begins the next conceptual step and is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration.

It remains to prove that (iii) *implies* (ii). Let

$$i\colon U \longrightarrow X$$

denote the canonical immersion of U into X. In view of the exact homology sequence associated with the closed subset Y (I, 2.11), one sees that (iii) is *equivalent to*:

(iv) $R^i i_*(\mathcal{F}|_U)$ is coherent for $i < n$.

French printed p. 94 Indeed, there is an exact sequence

$$0 \longrightarrow \mathcal{H}^0_Y(\mathcal{F}) \longrightarrow \mathcal{F} \longrightarrow i_*(\mathcal{F}|_U) \longrightarrow \mathcal{H}^1_Y(\mathcal{F}) \longrightarrow 0.$$

Now $\mathcal{H}^0_Y(\mathcal{F})$ is a quasi-coherent subsheaf of the coherent sheaf $\mathcal{F}$, and is therefore coherent. Hence $\mathcal{H}^1_Y(\mathcal{F})$ is coherent if and only if $i_*(\mathcal{F}|_U)$ is coherent. Moreover, for $p > 0$, the exact cohomology sequence associated with the closed subset Y reduces to isomorphisms

$$R^p i_*(\mathcal{F}|_U) \xrightarrow{\approx} \mathcal{H}^{p+1}_Y(\mathcal{F}).$$